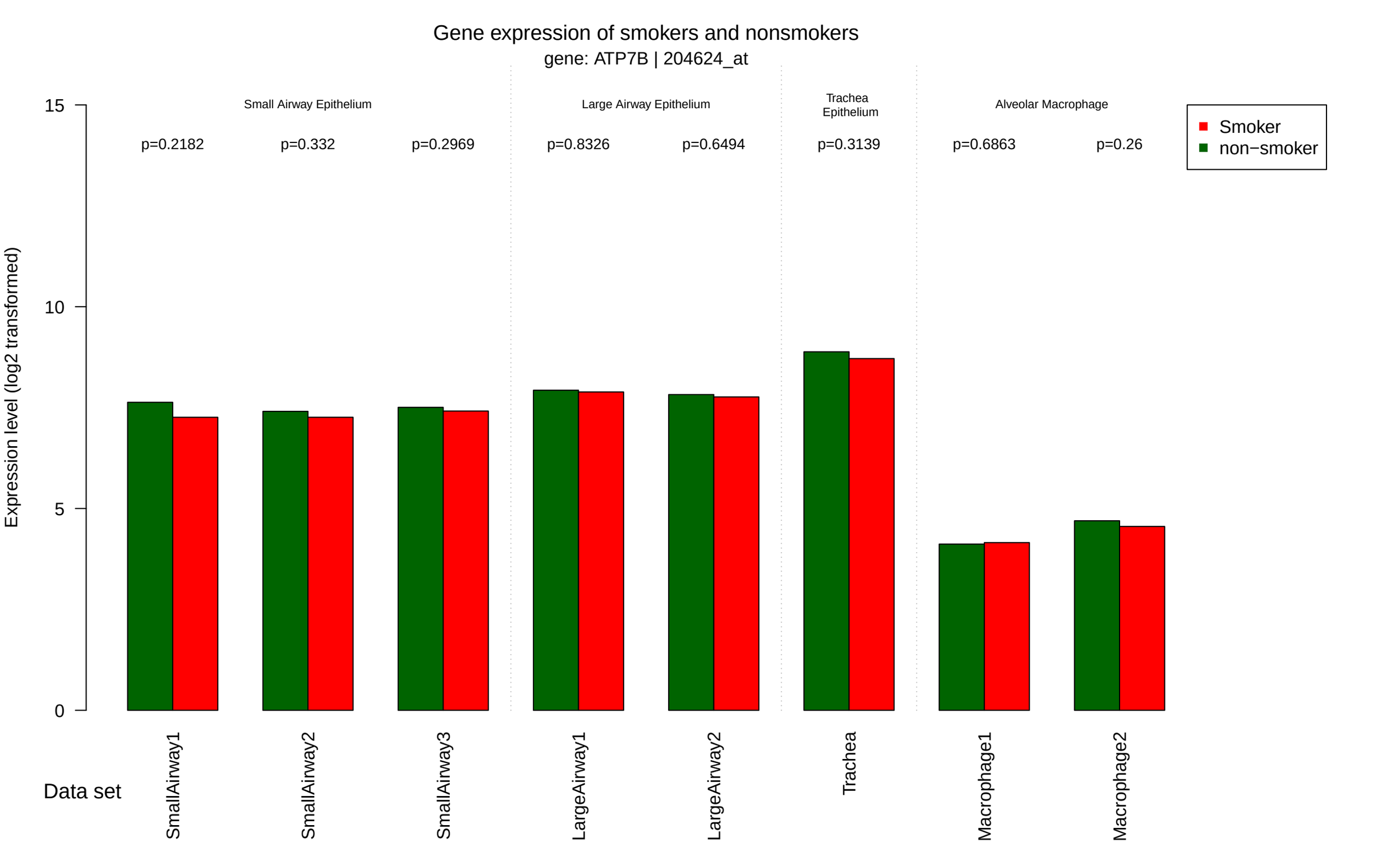
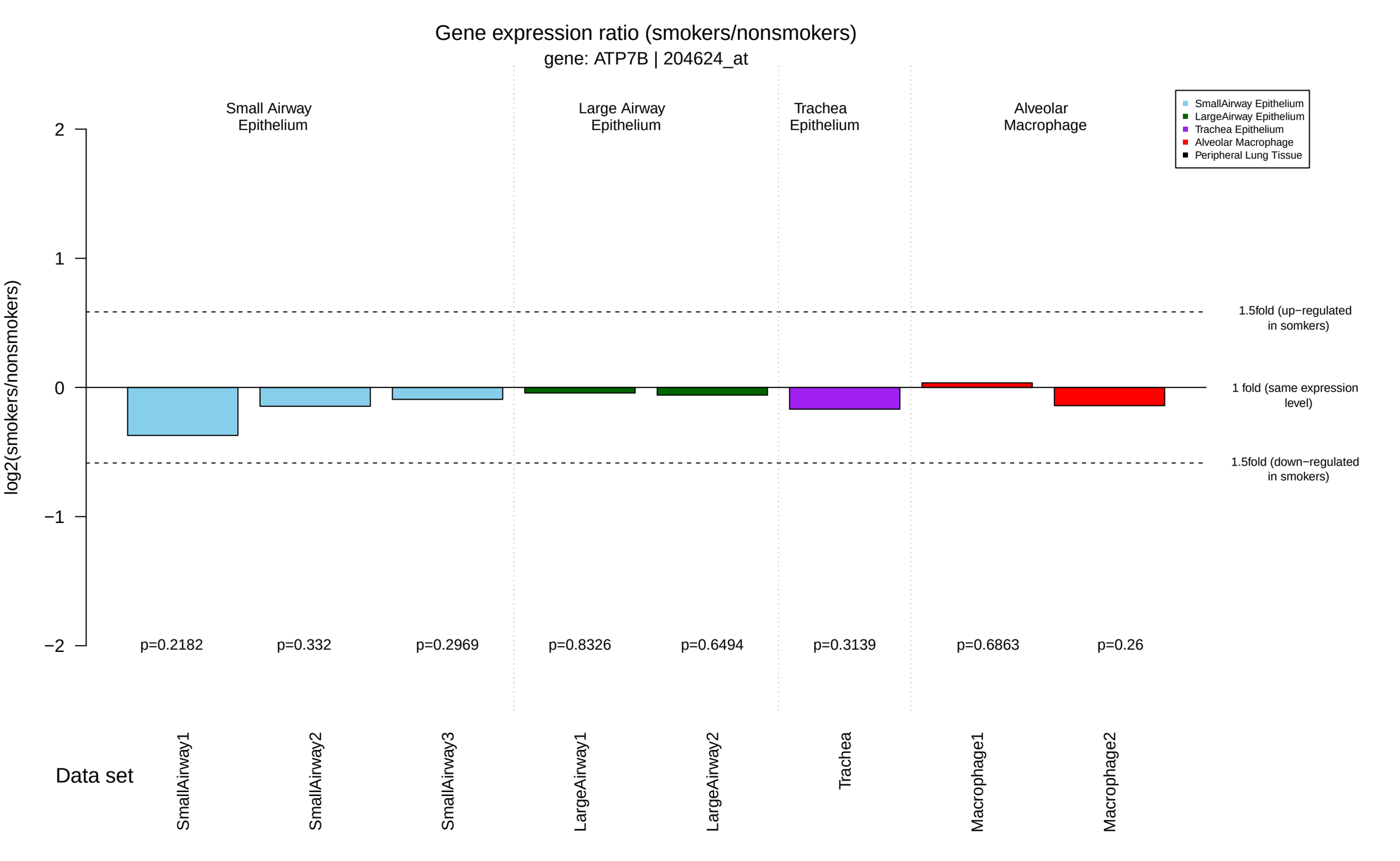
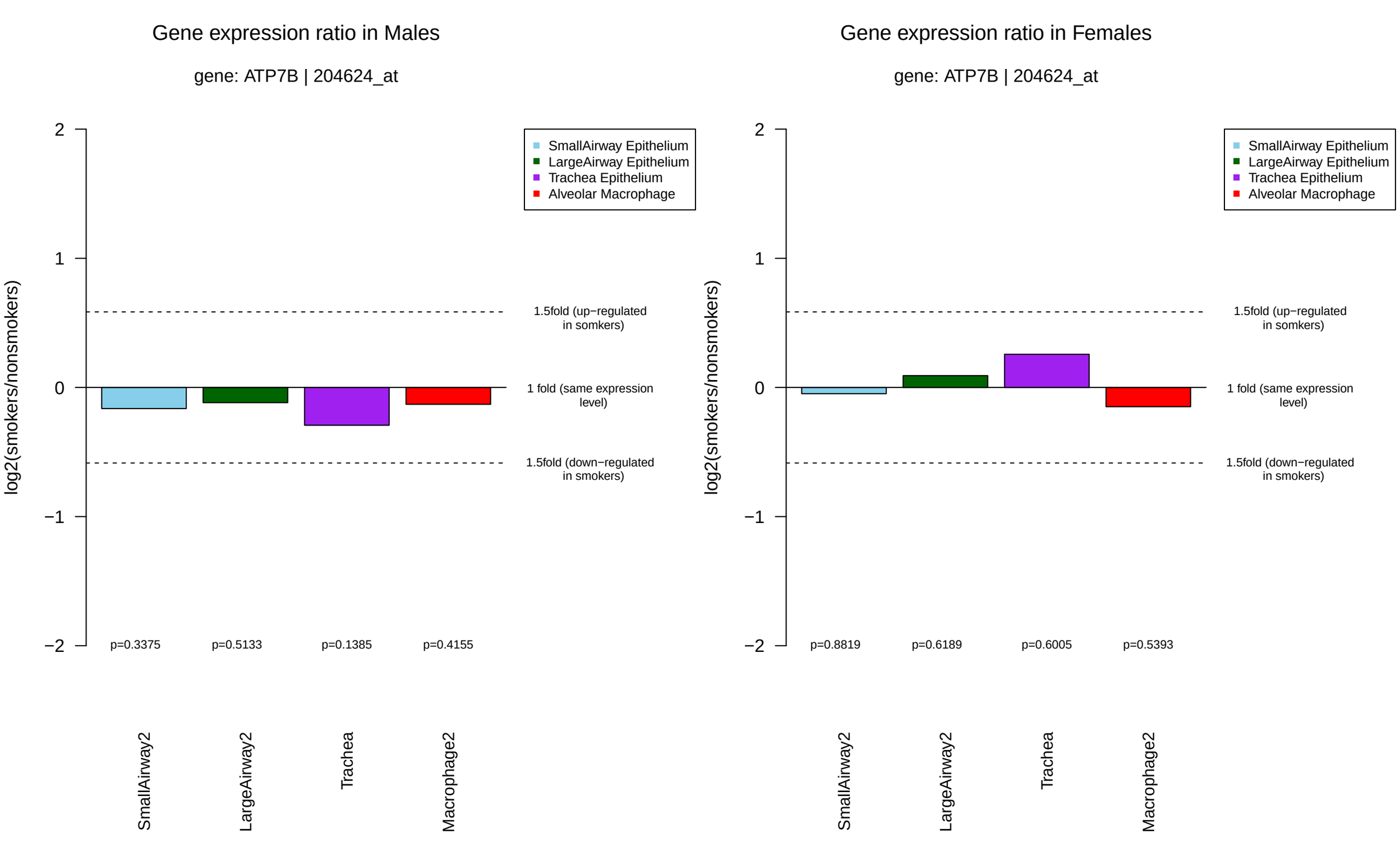
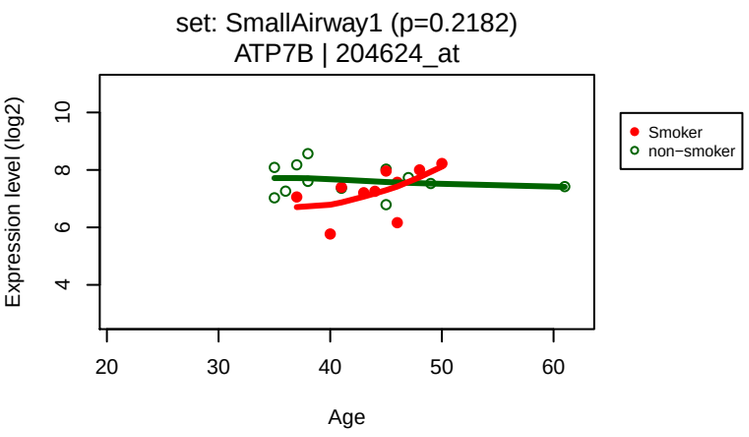
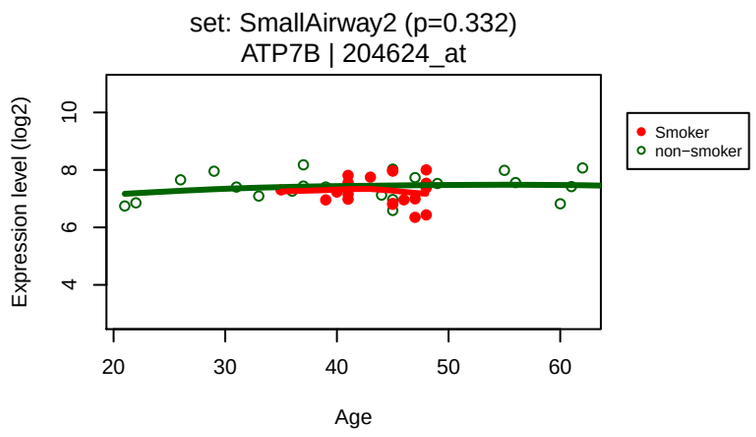
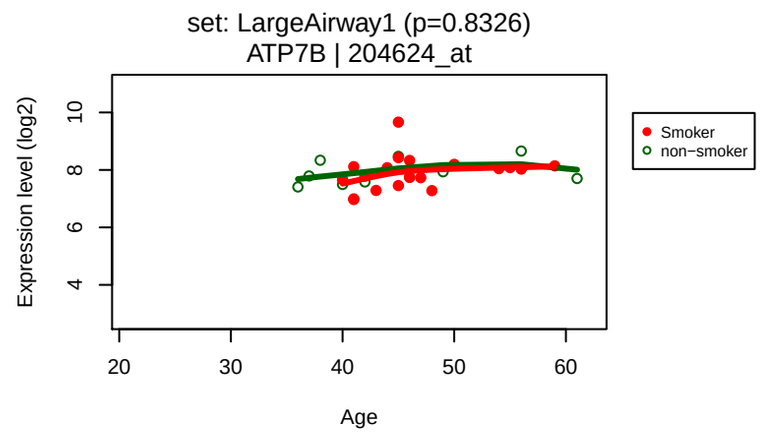

Gene expression of smokers and nonsmokers
gene: ATP7B | 204624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2182
p=0.332
p=0.2969
p=0.8326
p=0.6494
p=0.3139
p=0.6863
p=0.26
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

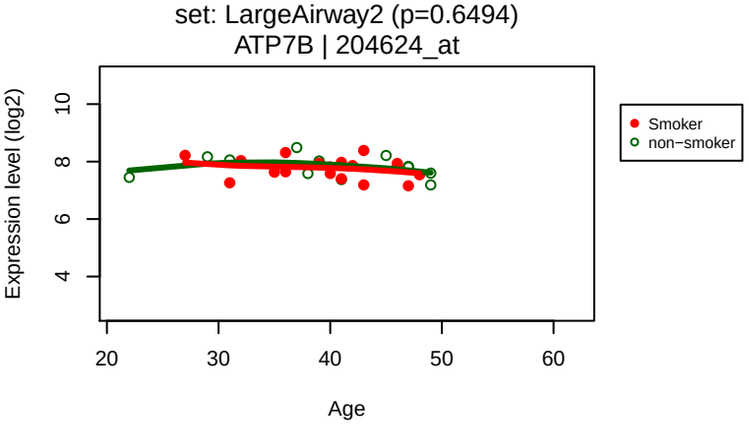

Gene expression ratio (smokers/nonsmokers)
gene: ATP7B | 204624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2182
p=0.332
p=0.2969
p=0.8326
p=0.6494
p=0.3139
p=0.6863
p=0.26
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

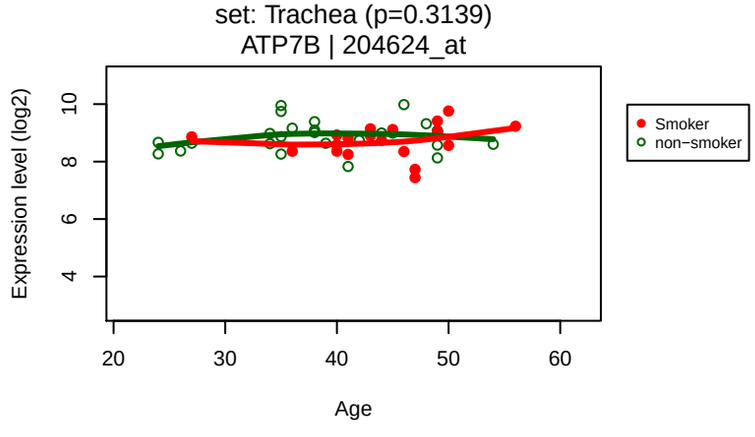

Gene expression ratio in Males
gene: ATP7B | 204624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3375
p=0.5133
p=0.1385
p=0.4155
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP7B | 204624_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8819
p=0.6189
p=0.6005
p=0.5393
SmallAirway2
LargeAirway2
Trachea
Macrophage2

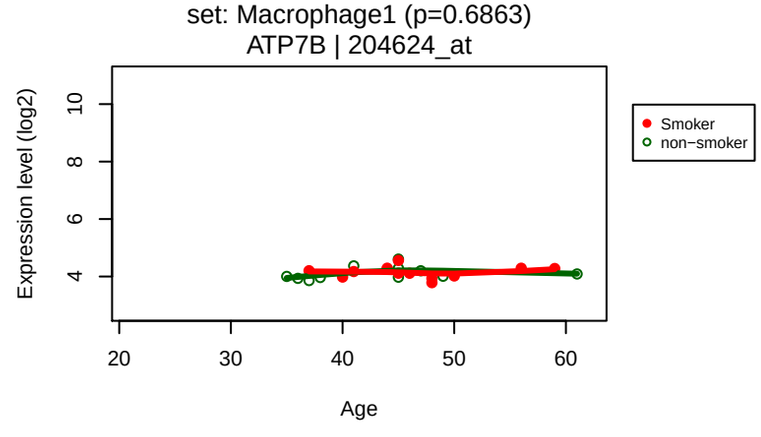

set: SmallAirway1 (p=0.2182)
ATP7B | 204624_at
Expression level (log2)
Age
Smoker
non−smoker

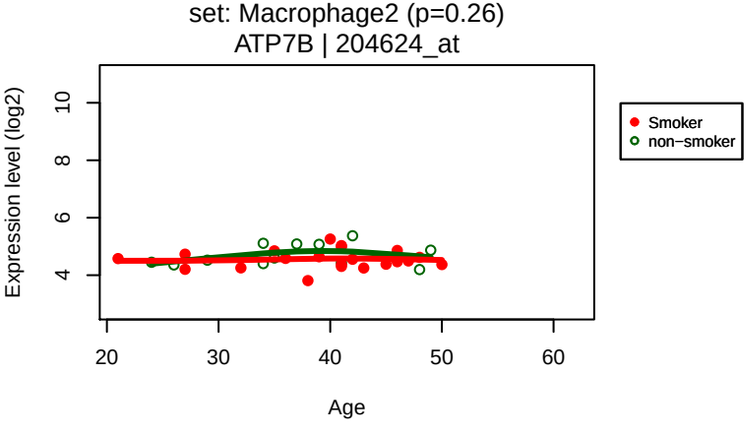

set: SmallAirway2 (p=0.332)
ATP7B | 204624_at
Expression level (log2)
Age
Smoker
non−smoker

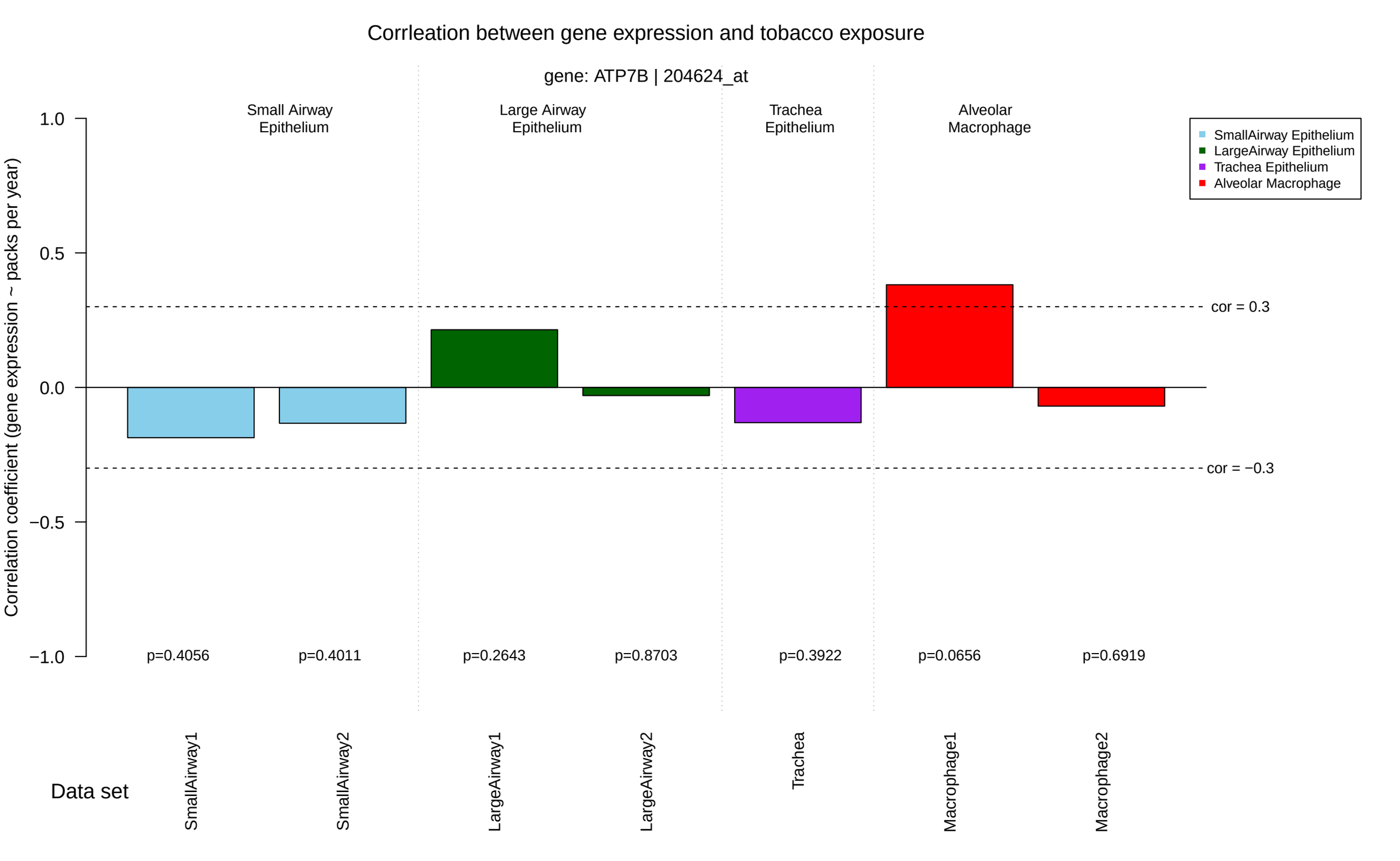

set: LargeAirway1 (p=0.8326)
ATP7B | 204624_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6494)
ATP7B | 204624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3139)
ATP7B | 204624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6863)
ATP7B | 204624_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.26)
ATP7B | 204624_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP7B | 204624_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4056
p=0.4011
p=0.2643
p=0.8703
p=0.3922
p=0.0656
p=0.6919
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2